

Drake Ferronyalka

U.S. Citizen | U.S. Navy Veteran | Active Secret Clearance (DoD) | dferronyalka@gmail.com | 941-920-4541 | linkedin.com/in/drakeferronyalka

PROFESSIONAL SUMMARY

Mechanical Engineering student and U.S. Navy veteran with an active DoD Secret Clearance and a track record of taking designs from concept through fabrication and validation. Hands-on experience across R&D prototype development, precision machining, and mechanical design for defense, research, and commercial applications. Proficient in SolidWorks (CSWA Certified), GD&T, DFM/DFA, and CNC manufacturing processes; disciplined approach to requirements, documentation, and procedure compliance.

EDUCATION

University of Central Florida B.S. Mechanical Engineering	Jan 2026 – Present
State College of Florida A.A. President's List (Fall 2024, Spring/Fall 2025); Dean's List (Spring/Summer 2024)	Dec 2025

EXPERIENCE

Machinist Apprentice <i>University of Central Florida, Office of Research</i>	May 2026 – Present <i>Orlando, FL</i>
- Fabricate components in direct support of Senior Design teams and faculty-led research labs, applying first-principles reasoning to translate design intent into producible geometry under real shop constraints. - Produce GD&T drawings and SolidWorks designs for shop fixtures and CNC tooling used across multiple concurrent research and student projects. - Advise students and research teams on design-for-manufacturability, reducing rework and improving part producibility on available shop equipment.	
Mechanical Design and Fabrication Technician <i>AFNY Specialties</i>	2020 – 2025 <i>Ellenton, FL</i>
- Designed and fabricated SolidWorks assemblies ranging from single-piece brackets to 10-15 part mounts and suspension hardware engineered for high-vibration duty; produced BOMs and assembly notes supporting fabrication and installation. - Standardized repeatable threaded features using a FlexArm tapping system, optimizing tap geometry, torque control, and lubrication practices to reduce tool breakage and maintain thread quality.	
Fire Controlman / 400 Hz Power System Maintainer <i>United States Navy</i>	2017 – 2023 <i>Various Locations</i>
- Maintained and troubleshooted MK 99 T-1348 Radar Transmitter and ACSSFC 400 Hz power systems aboard DDG-class ships, executing preventive and corrective maintenance to sustain operational readiness in high-tempo environments. - Led and trained 10+ technicians on diagnostics, troubleshooting workflows, and procedure compliance; standardized checkouts and technical direction improved team execution.	

PROJECTS

DTA-1 Single-Wheel EV Drivetrain Prototype <i>VESC Motor Control, Custom Planetary Gearbox Design, Embedded Systems Integration</i>	Jun 2026 – Present
- Evaluating BLDC motor, VESC controller, and sensor/encoder candidates for integration into a single-wheel electric drivetrain, weighing torque, thermal, and duty-cycle requirements to guide selection. - Designing a custom two-stage 10:1 planetary gearbox from load and duty-cycle requirements, structuring the project through a formal requirements matrix, BOM, and phase plan ahead of procurement and bench testing. - Scoping data acquisition and supervisory control architecture via Raspberry Pi to integrate electrical and mechanical subsystems into a testable prototype.	
Wheel Light Mounting System (Patent Pending) <i>SolidWorks, GD&T, Prototype Fabrication</i>	
- Designed a modular mounting architecture for off-road lighting with emphasis on vibration resistance, packaging, and serviceability; produced manufacturing-ready drawings and BOM. - Fabricated prototypes via CNC plasma and secondary operations; validated performance via torque checks and road testing.	

SKILLS

Design Practices: GD&T, DFM/DFA, Tolerance/Fit, BOMs, Design Iteration
Manufacturing: CNC plasma cutting, Milling, Turning, Sheet metal/plate fabrication, Press brake forming, Tapping/threaded features, Assembly and integration
CAD/Design: Dassault Systemes SolidWorks (CSWA Certified), Siemens NX (modeling)
Tools: MATLAB, Microsoft Excel, Word, PowerPoint
Programming: Python, C

CLEARANCE

- Active Secret Clearance (DoD) | Adjudicated Dec 2017